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CADBOARD

Vision System For Cardboard Feature Inspection

Author:

Oskar Haapalo
Karim Yasser

Student ID:

Oskar:11109666
Karim:11063520

Advisor:

Prof. Marco Tarabini

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1 | Introduction

Accurate inspection of die-cut features, such as cuts, holes, and creases, is critical for maintaining the structural and functional quality of cardboard packaging. Manual inspection is labor-intensive and prone to inconsistency. This project presents a camera-based visual inspection system designed for offline quality control stations. The system compares captured images of cardboard samples against their CAD specifications to verify feature accuracy. It supports both live image capture via an industrial IDS camera and analysis of pre-captured images. Using a combination of computer vision techniques and geometric matching, the system overlays and aligns a perspective-corrected CAD layout onto the image, allowing precise evaluation of deviations in cut and crease placement.

Our approach offers several key advantages over traditional manual inspection and purely data-driven AI methods:

- **Direct CAD-Based Validation:** The system compares captured images of cardboard samples directly against their CAD specifications. This ensures true-to-design accuracy rather than comparing to a "golden sample" image.
- **High Robustness to Placement & Perspective:** The alignment pipeline, particularly the iterative homography module, automatically compensates for non-ideal viewing angles and distortion, effectively bypassing the need for explicit geometric calibration or perfectly controlled placement.
- **Explainable & Deterministic Results:** Unlike "black box" AI models, every defect is detected through a clear, rule-based process, such as boolean mask comparisons for material conformance or quantitative fill-factor analysis for crease integrity.
- **No Training Data Required:** The system can inspect a new part design immediately upon loading its DXF file. It does not require large, pre-existing datasets or lengthy, model-specific training sessions.

2 | Assumptions and Intended Use Environment

The system targets a stationary quality control (QC) station, where samples of cardboard are inspected in a controlled environment. Images are captured with a fixed camera setup after placement of the cardboard on a designated surface.

2.1. Expected Use Environment

Key environmental characteristics assumed for this use case:

- **Consistent lighting:** The inspection area is assumed to maintain stable lighting conditions with a high-contrast background. To maximize the visibility of creases and cut lines, the lab setup leverages the principles of *raking light illumination*. This technique uses directional lights at an oblique angle to cast sharp shadows from topographical features, significantly enhancing their contrast. In our setup, this was achieved using white LED light bars; positioning them at an angle of $80 - 90^\circ$ relative to the camera axis notably improved the visibility of crease lines. While this creates optimal conditions for HSV-based segmentation, the system includes tunable parameters to adapt to moderate lighting variations. For industrial applications, enclosed lighting setups, such as those illustrated in Figure 2.1 described by Leotta et al. [1], can be adopted to eliminate ambient interference and ensure consistent image quality across inspections.

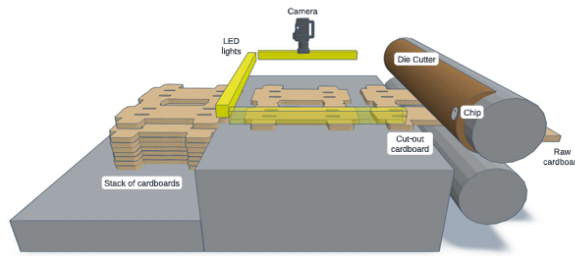


Figure 2.1: Illustration of optimal lightning setup in production line

- **Fixed camera geometry:** The camera is rigidly mounted with a fixed focal length and oriented orthogonally to the inspection surface. Ideally, lens distortion would be corrected using standard checkerboard calibration methods (e.g., via OpenCV), which can be performed on site. However, in our implementation, we compensate for distortion by directly aligning the CAD DXF features to the captured image using a mean squared error (MSE) minimization approach, effectively bypassing the need for explicit geometric calibration and controlled placement.
- **Layered DXF input:** CAD files are expected to follow a structured convention

with distinct layers for outlines, cut-outs, and crease lines, facilitating layer-wise parsing and alignment.

- **Backside inspection assumption:** All inspections are assumed to be performed on the backside of the cardboard, i.e., the unprinted, brown surface. This ensures consistent texture and color for segmentation and minimizes interference from printed graphics, glossy coatings, or labels that may be present on the front side.

3 | System Overview

This section outlines the hardware components and software infrastructure used to implement the inspection system.

3.1. Hardware Setup

- **Camera:** IDS UI-3860CP-C-HQ Rev.2 (AB00853)
 - **Lens (Lab):** F1.4/8mm — chosen for sharp focus and a wide field of view, suitable for inspecting various cardboard sizes. The focus was manually tuned during setup.
 - **Mounting:** The camera was mounted orthogonally to the inspection surface to ensure consistent geometry and minimize perspective distortion.
 - Industrial Setup Recommendations:
 - * **Aperture:** Set between f/4 to f/8. Lower f-numbers (e.g., f/2.8) increase brightness but reduce depth of field. Higher f-numbers (e.g., f/8) improve sharpness and ensure full sample focus. A mid-range value like f/5.6 offers a practical balance between exposure and focus.
 - * **Focal Length:** Recommended range is 6–12mm. Shorter focal lengths offer a wider field of view but may introduce distortion. Longer focal lengths reduce distortion and improve image clarity, especially near the edges. This is highly dependent on mounting ability/position of the camera in the setup.
- **Lighting configuration:**
 - Red light was available but did not yield a consistent improvement in contrast or edge detection in our approach.
 - Experiments were conducted using white LED light bars positioned at various angles. Notably, placing the bars at an angle of 80–90° relative to the camera axis significantly improved the visibility of crease lines. This effect aligns with the principles of *raking light* illumination, a technique commonly used in visual inspection to enhance surface textures and reveal fine details.
- **Background:** A matte black canvas was used to emulate an industrial inspection surface or a conveyor belt.

3.2. Software Stack

- Language: Python 3.13
- GUI Framework: Tkinter
- Libraries: OpenCV, NumPy, SciPy (KDTree), ezdxf, PIL, pyueye

4 | Workflow Description

The application is designed as a modular pipeline that cleanly separates key tasks: image acquisition, CAD parsing, alignment, and visual inspection. In its current form, the system is tailored for use in an offline quality control (QC) station, where operators manually place cardboard samples for inspection.

While it supports both live image capture via the IDS industrial camera and analysis of pre-captured files, the main use case centers on controlled, manual inspection rather than continuous inline operation. This chapter describes each stage of the pipeline in detail, beginning with how images are acquired and prepared for processing.

4.1. Image Acquisition

Users can choose between two methods:

- **Live Capture:** Directly from the IDS UI-3860CP camera. The system sets the color mode to BGR8, allocates memory, and captures an image with resolution matching the sensor's maximum (1936x1096 pixels).
- **Manual Load:** Load a saved image (JPG, PNG, BMP, etc.) from disk.

Captured images are displayed in the GUI and stored in memory for further processing. The success or failure of the operation is reflected in GUI status indicators

4.2. CAD Overlay Import

The user can load a DXF file containing CAD definitions of the expected cut and crease patterns. The system supports three specific layers:

- **OUTLINE:** Exterior profile of the cardboard
- **HOLES:** Internal cutouts
- **CREASES:** Fold lines

We read each layer using the ezdxf library and break down the shapes into simple line segments. Since some designs come in many small pieces, we use a custom method to reconnect lines that are close together into full shapes. This makes it easier to handle more detailed or complex drawings.

Once this is done, the features are saved in a clear, organized format and shown on screen as both numbers and drawings so the user can double-check that everything was read correctly.

5 | Alignment & Inspection Pipeline

With images and CAD features available, the next step is to align them precisely. The alignment and inspection pipeline described below.

5.1. Contour Extraction and Preprocessing

The image is first converted from BGR to HSV color space, and a fixed HSV range is used to isolate the cardboard sheet from the background. External contours are then extracted using `cv2.findContours`. The largest contour is assumed to be the cardboard boundary and is used as the reference shape for alignment.

CAD outlines, parsed from the DXF file, are flipped vertically to match the image coordinate system. This transformation accounts for the inverted Y-axis used in most image display conventions.

5.2. Affine Alignment (Spatial Moments + ICP)

The initial alignment uses spatial moments to estimate translation, scale, and rotation parameters that map the CAD contour onto the image contour. The steps include:

1. Computing image and CAD centroids
2. Estimating principal orientations from second-order moments
3. Estimating uniform scale based on area
4. ICP (Iterative Closest Point) refines alignment, by iteratively minimizing the mean squared distance between CAD and image contours using nearest-neighbor matching.

Finally, because the CAD's coordinate system may or may not be different from the images, sometimes leading to an inverted alignment. The solution is as follows,

Two passes are executed:

- Pass 1 (Y-Flipped CAD): Matches flipped CAD contour to image
- Pass 2 (Original CAD): Matches original CAD contour

The transformation with lower MSE is selected for final use.

5.3. Homography Refinement

If enabled, a perspective transformation is further refined using corner detection and iterative homography updates. This procedure:

- Extracts polygonal corners from both contours
- Matches them using nearest-neighbor search (KDTree)
- Estimates a homography matrix using RANSAC

An iterative loop runs until the MSE ceases to improve, allowing for automatic convergence. If the homography outperforms the affine method in terms of alignment error, it is selected. Otherwise, the affine result is retained.

5.4. Overlay and Visual Inspection

Once alignment is completed, all features (outline, holes, and creases) are transformed into image coordinates using the best-fit transformation. They are drawn over the original image using color-coded lines:

- Yellow: Outline
- Magenta: Holes
- Cyan: Creases

The result is displayed within the GUI, allowing users to visually confirm alignment quality and detect deviations.

5.5. Anomaly Detection and Inspection

After a high-fidelity alignment is achieved, the system performs a programmatic inspection to identify and quantify defects. This stage uses a combination of morphological image processing, boolean mask comparisons, and feature-specific analysis algorithms to ensure the physical part conforms to its digital specification. The inspection is broken down into the following distinct procedures:

5.5.1. Material Conformance Analysis

This procedure detects deviations in the part's overall silhouette by comparing the ideal shape from the CAD model against the actual shape detected in the image.

1. **Mask Generation:** Two primary binary masks are created. The "Ideal Mask" is generated by drawing the filled, aligned CAD outline. The "Real Mask" is generated from the contours of the physical part detected in the camera image.
2. **Extra Material Detection:** To find extra material, the Ideal Mask is first expanded using a morphological 'dilate' operation. The size of this expansion is controlled by a user-defined tolerance, creating an upper-bound for the acceptable shape. This expanded mask is then subtracted from the Real Mask. Any remaining non-zero pixel regions in the resulting image correspond to areas of excess material (e.g., burrs or flash).
3. **Missing Material Detection:** Conversely, to find missing material, the Ideal Mask is shrunk using a morphological 'erode' operation, creating a lower-bound tolerance. The Real Mask is then subtracted from this shrunk mask. The resulting image highlights all areas where the physical part is missing material that should be present according to the CAD specification. This 'missing material mask' is passed to the hole verification stage.

5.5.2. Hole Occlusion Verification

This check uses the result from the missing material detection to verify the integrity of internal features.

- **Region-Based Analysis:** The algorithm iterates through each hole feature defined in the CAD model. For a correctly formed, open hole, the system expects the corresponding region within the missing material mask to be fully active (i.e., white).
- **Occlusion Ratio Calculation:** The system measures the actual area of "missing material" detected within the bounds of each expected hole. An 'occlusion ratio' is calculated as $1 - (\text{Detected Missing Area} / \text{Expected Hole Area})$. A ratio of 0 indicates a perfectly clear hole, while a ratio of 1 indicates a completely blocked hole.
- **Thresholding:** If the calculated occlusion ratio for any hole exceeds the user-defined tolerance, it is flagged as an occluded defect. Any remaining regions from the 'missing material mask' that do not correspond to a defined hole are flagged as separate "missing material" defects.

5.5.3. Crease Integrity Assessment

Because creases are textural features rather than shape boundaries, their inspection requires a different methodology based on local image intensity.

- **Adaptive Thresholding:** Instead of a single global threshold, a `cv2.adaptiveThreshold` is applied to the grayscale image. This method calculates a unique threshold for small, localized pixel neighborhoods, making it highly effective at highlighting dark linear features like creases, even under non-uniform lighting conditions. This produces a clean binary mask of all potential creases.
- **Corridor Analysis and Fill Factor:** For each expected crease from the CAD model, a "corridor" mask is generated along its aligned path. The system then performs a bitwise AND operation between this corridor and the adaptive threshold mask. A "fill factor" is calculated by dividing the number of active pixels in the result by the total number of pixels in the corridor. This score, from 0% to 100%, represents how completely the detected line fills the expected path.
- **Classification:** This quantitative fill factor is compared against a user-defined threshold to classify the crease as either present and correct, or defective due to being faint, incomplete, or missing entirely.

6 | Evaluation

6.1. Evaluation Metrics

To assess the systems inspection accuracy, we focus on the following metrics:

- **Precision:** Defined by the mean squared error (MSE) in pixels between the aligned CAD overlay and the actual image contours. This metric reflects how accurately the system matches the expected geometry to the physical sample. A low MSE indicates precise overlay, which is a critical prerequisite for effective defect detection.

Once the CAD overlay is well-aligned, the system evaluates defects by comparing binary masks of expected and observed features. These include differences in overall shape (extra or missing material), blocked holes, and underdeveloped crease lines. Thus, alignment precision directly impacts the reliability of downstream anomaly detection.

- **Execution Time:** The time required to complete alignment and inspection for a given sample. This metric reflects the systems responsiveness and suitability for a real-time or semi automated workflow (if the system would be implemented in line on a convener belt set up). Measured using standard time-stamping in the code.
- **Robustness:** Can the system maintain successful alignment and accurate inspection under minor image distortions, such as changes in lightning, contrast or perspective shifts. Evaluated via visual examples presented below.

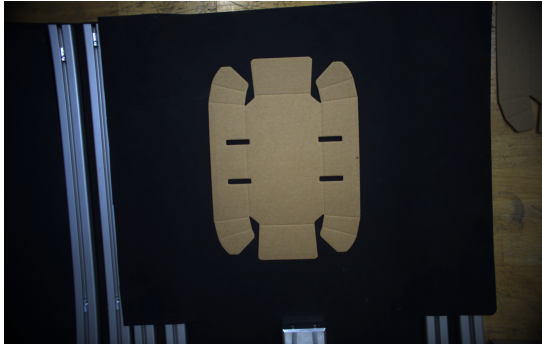
6.2. Evaluation Procedure

6.2.1. Robustness Validation

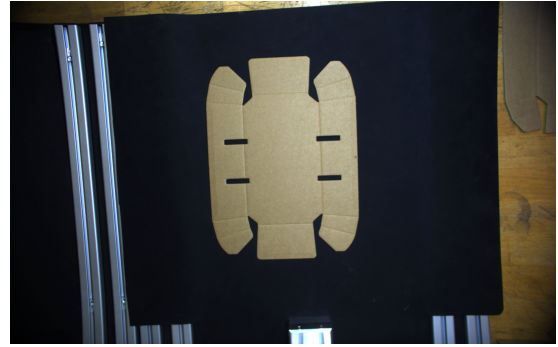
Testing multiple lighting configurations using the same cardboard sample within a controlled lab environment allowed us to identify setups that produced the most accurate alignment results, the results are summarized in Table 6.1. In particular, the lowest MSE values were observed when the LED light bar was positioned directly above the sample at approximately camera height, or at an oblique angle (80–90°) relative to the long edge of the cardboard. It is important to note that the high anomaly count in Test 3 is not indicative of actual manufacturing defects. Rather, it reflects the algorithm’s sensitivity to overexposed regions, which were mistakenly flagged as potential defects due to strong reflections or glare.

ID	Lighting Condition	Affine MSE	Homography MSE	Anomalies	Best Method
1	Above	16.15	3.84	4	Homography
2	Above-Closer	18.12	5.00	52	Homography
3	Long Side	15.21	3.15	461	Homography
4	Short Side	16.48	5.31	32	Homography
5	Red Light	15.31	3.38	24	Homography

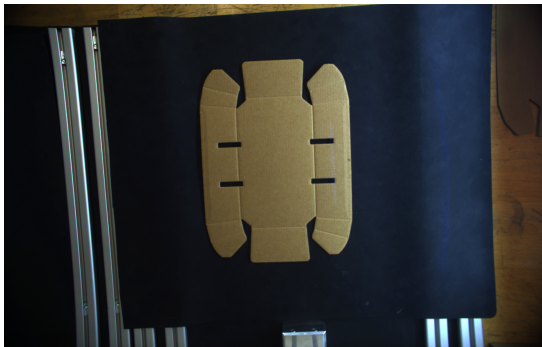
Table 6.1: Alignment performance across different lighting conditions. Homography consistently yields lower MSE than the baseline affine method.



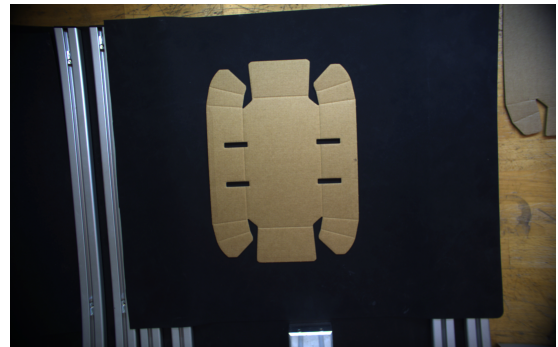
(1) Above



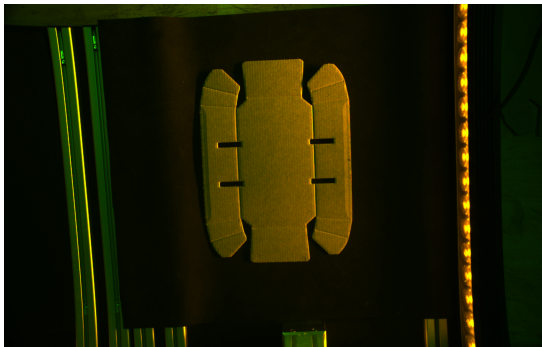
(2) Above-Closer



(3) Long Side



(4) Short Side



(5) Red Light

Figure 6.1: Different lightning setups used to test robustness and evaluate light bar placement.

6.2.2. Perspective Correction Evaluation

To assess the system's ability to handle perspective distortion, a benchmark test was conducted using an angled image of a cardboard sample. Initial alignment via affine transformation yielded a high mean squared error (MSE) of 294.87, indicating poor fit due to the strong perspective skew. Upon activating the iterative homography refinement

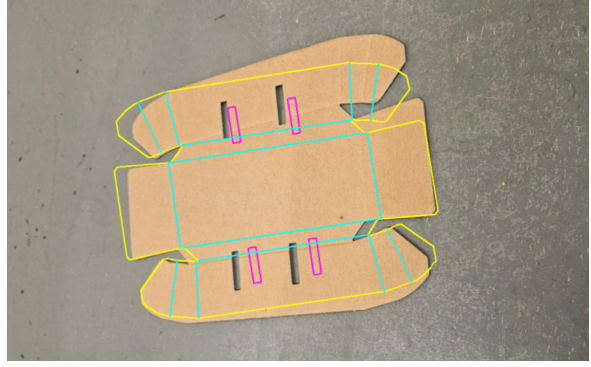
module, the system significantly improved alignment, achieving a final MSE of 11.57.

As shown below (Figures 6.2a, 6.2b, 6.2c, and 6.2d), the homography-corrected features align tightly with the cardboard edges and crease lines, enabling accurate visual inspection and reliable anomaly detection. The final defect analysis correctly identified 2 out of 14 crease segments as potential deviations, consistent with expectations.

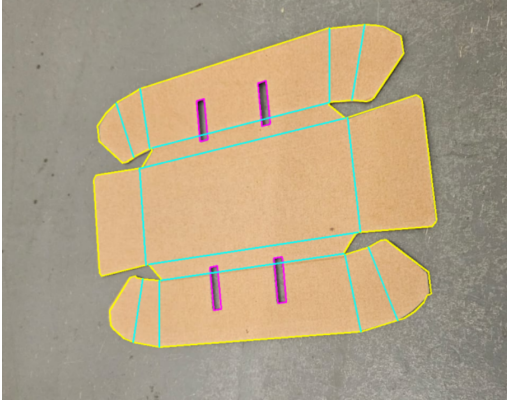
This confirms the effectiveness of the homography refinement in the system to accurately align under non-ideal viewing angles a critical feature for real-world applications where perfect camera orientation cannot be guaranteed.



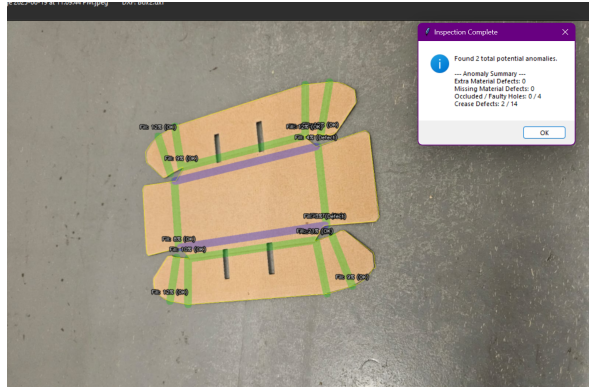
(a) Original perspective image



(b) Initial affine overlay



(c) Refined homography overlay



(d) Defect detection result

Figure 6.2: Alignment results from the perspective correction test.

6.2.3. Execution Time

Presented below in Table 6.2 is the summarized runtime performance across the systems primary processing stages. While the affine alignment method is faster, averaging 0.044s, its lower accuracy (Table 6.1) makes it less suitable for fine-detail inspection. The homography-based method takes slightly longer (avg. 0.073s), but consistently achieves

better MSE, making it the preferred option when precision is critical. In use cases where processing time is more constrained than geometric accuracy (e.g., high-speed production lines), affine alignment may still be acceptable. Anomaly and defect detection, involving multiple contour passes and geometry checks is naturally the most time-consuming stage averaging 0.19s. Altogether the full pipeline executes in approximately 0.26s per sample, suggesting that the system is well-suited for near-real-time inspection tasks, particularly in offline or semi-automated QC scenarios.

Performance Metric	Minimum Time (s)	Maximum Time (s)	Average Time (s)	Samples
1. Contour Detection (for Alignment)	0.0124	0.0626	0.0399	14
2a. Full Alignment (Affine)	0.0366	0.0516	0.0441	2
2b. Full Alignment (Homography)	0.0382	0.1071	0.0731	12
3. Anomaly & Defect Analysis	0.1382	0.3036	0.1910	9
Total Estimated Processing Time	~0.17	~0.41	~0.26	-

Table 6.2: Timing benchmarks for key stages of the inspection pipeline.

6.3. Conclusion

The system demonstrates strong performance for inspecting die-cut features on cardboard in controlled quality control (QC) environments. Through robust alignment methods, particularly homography refinement, the system achieves high geometric precision, evaluated through MSE alignment testing.

A key strength of the approach lies in its visual overlay pixel-level inspection feature. By comparing masks derived from the aligned CAD model to observed features in the image, the system effectively detects anomalies such as extra or missing material, occluded holes, and incomplete crease lines. The low alignment error achieved in perspective-corrected tests, combined with consistent detection of actual defects, confirms the reliability of this strategy under moderate distortion or illumination variability.

Despite these strengths, some limitations persist. Rolling shutter distortion (with the current camera), especially during movement, if adapted in line, may impact contour extraction. Furthermore, the system currently assumes clean, and unprinted cardboard surfaces. Extending support to printed, reflective, or warped materials would require preprocessing development.

Overall, the system’s speed, modularity, and analysis make it a practical and accurate tool for semi-automated offline inspection. Future work may explore full in-line integration, improved crease detection under noisy lighting, and data-driven methods for more generalized defect classification.

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- [1] M. Leotta, A. Bontempelli, S. Turchi, and B. Caputo. VISION: Vision-based Inspection System for Industrial Object Normality. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024. URL https://www.diag.uniroma1.it/leotta/publications/VISION_ECCV2024.pdf.